Simple Linear Regression, Example

CSU Chico, Math 314

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Hypothesis

Load the built-in `mtcars` dataset and use linear regression to answer the question: does a car's weight (wt) predict its gas mileage (mpg)?

Getting Started

Load some libraries.

```
suppressMessages({library(ggplot2)
      library(dplyr)})
```

Tips to Keep in Mind

- Plot the data (be sure to include geom_smooth(), too)
- Create your linear model with lm()
- Check your assumptions using the residuals (with rstandard()) and fitted values (with fitted())
- Use the summary results to determine your important variables and values
- Reject or fail to reject the null hypothesis (that the slope is 0)? Consider $\alpha = 0.05$.
- Use confint() to get a 95% confidence interval for the slope.

Plot the data! (Framework)

```r
p <- ggplot(data, aes(x, y)) +
    stat_smooth(method="lm")
```

Fit Linear Regression

```r
fit <- lm(y ~ x, data=data)
## then check assumptions
```

Check Assumptions

Once you define the standardized residuals (e) and fitted values (yhat), be sure you place them in a dataframe object:

```r
df <- data.frame(e,yhat)
```

Linearity, Constant Variance, Normality

Be sure to complete both plots for these to make your conclusions.

Standard Output

When using summary() on the linear model object, be sure you can identify coefficients, test statistics, and p-values for slope and intercept, as well as the adjusted R-squared values.

Interpretations

What does the confidence interval represent for the slope in this case?

Try This One

Repeat these steps, but now determine whether horsepower (hp) can be used to predict mileage (mpg). Do you reject or fail to reject the hypothesis that the least squares line's slope is 0?

references I

Hadley Wickham. *ggplot2: elegant graphics for data analysis.* Springer Science and Business Media, 2009.

Adapted from notes by Dr. Roualdes.